

Tuhin Garai

📞 +919845779355 ✉ tuhingarai123@gmail.com 📍 Bengaluru, Karnataka, India 🔗 tuhingarai.in 🌐 linkedin.com/in/tuhingarai
🐙 github.com/nightcodex7 📝 medium.com/@nightcode_x

🧠 SKILLS

Programming/Scripting: Python, Bash, Shell Scripting

DevOps & Cloud: Docker, Docker Compose, Kubernetes, Jenkins, CI/CD, AWS (EC2, VPC, S3, IAM, SNS, CloudWatch), GitHub Actions, Ansible, Terraform

Tools & Platforms: Git, GitHub, Linux

Systems & Administration: Linux Administration, System Monitoring, Troubleshooting, Log Analysis

📁 PROJECTS

Serverless EC2 Monitor & Control Dashboard

- Built a serverless EC2 monitoring and automation system using AWS Lambda, API Gateway, DynamoDB, and S3; published findings in IJRSET.
- Automated EC2 instance lifecycle management using usage-based triggers, reducing manual monitoring and shutdown effort by ~1 hour daily.
- Configured AWS-based monitoring, alerting, and scalable infrastructure workflows to improve observability, reliability, response time, and operational availability.
- Documented deployment architecture, workflows, and operational processes.

Lenovo Vantage Unofficial Linux Application

- Developed an unofficial Linux application replicating core Lenovo Vantage features for Lenovo laptops.
- Implemented battery conservation mode, rapid charging controls, and hardware management through Linux system interfaces.
- Authored installation and troubleshooting documentation supporting Fedora KDE users.

Zero Downtime Application Deployment using Kubernetes & AWS

- Designed a highly available cloud-native deployment architecture on Amazon EKS with multi-region failover using Route 53 health checks.
- Implemented Kubernetes rolling updates, readiness/liveness probes, and Horizontal Pod Autoscaler (HPA) for achieving zero-downtime deployments and self-healing recovery across simulated high-traffic scenarios.
- Integrated Prometheus and Grafana dashboards alongside CloudWatch for end-to-end observability of system health and scaling behavior.
- Configured warm standby disaster recovery with automated failover, reducing projected RTO by ~40% compared to cold standby baseline.

WARP + NextDNS Manager

- Developed a cross-platform CLI application automating Cloudflare WARP, WireGuard, and NextDNS setup, configuration, and service management.
- Implemented interactive setup, real-time status monitoring, diagnostics, and automated configuration workflows for Linux and Windows.

🏆 ACHIEVEMENTS

Android Open Source (AOSP) Community Contributor

- Tested and validated custom Android ROMs (crDroid, LineageOS, AOSPA, VoltageOS, and others) across 5+ devices including Nothing Phone 2, Xiaomi, and Poco series.
- Identified and reported bugs, system instability, and performance bottlenecks; captured diagnostic logs (logcat) to assist developers in debugging and optimization.
- Performed device flashing, recovery, and restoration via fastboot, TWRP, OrangeFox, and EDL mode across multiple device architectures.
- Worked with Magisk, KernelSU, and LSPosed for root configuration and system-level customization testing.

📖 PUBLICATIONS

Research Paper: IJRSET — Build a Serverless EC2 Monitor & Control Dashboard

📜 CERTIFICATES

Ethical Hacking & Information Security

National Institute of Electronics & Information Technology (NIELIT), Gorakhpur (2021)

Introduction to Industry 4.0 and Industrial IoT

National Programme on Technology Enhanced Learning (NPTEL), IIT Kharagpur (2022)

🎓 EDUCATION

Bachelors in Computer Application (BCA), Chandigarh University

2021 – 2024 | Mohali, Punjab

Masters in Computer Application (MCA), Jain University

Specialization: Storage & Cloud Technology (Major), AI & ML (Minor)

2024 – 2026 | Bengaluru, Karnataka